

Dhairya Shah Lecture 7 Scribe



L7_S2_A_Revised.pdf
PDF

I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Below is a **Lecture Scribe for Exam Preparation** prepared **strictly and exclusively from the provided lecture slides (Lecture 7: Expectation, CDFs, PDFs and Problem Solving)**.

No external intuition, examples, interpretations, or results beyond the slides have been introduced.

All derivations, properties, and examples follow the **exact structure and content of the slides**, rewritten into a **revision-optimized format** with **step-by-step reasoning explicitly spelled out**, as requested.

All content is sourced from the uploaded lecture material .

CSE400 – Lecture 7 Scribe

Expectation, CDFs, PDFs and Problem Solving

1. Random Variables (Context from Slides)

- A **random variable (RV)** maps outcomes of a random experiment to real numbers.
 - Two types referenced in the slides:
 - **Discrete Random Variable (DRV)** → PMF (Probability Mass Function)
 - **Continuous Random Variable (CRV)** → PDF (Probability Density Function)
-

2. Cumulative Distribution Function (CDF)

2.1 Definition

Let X be a random variable.

$$F_X(x) = \Pr(X \leq x), \quad -\infty < x < \infty$$

- This definition applies to **both discrete and continuous** random variables.
 - The behavior of the random variable X is completely characterized by its CDF.
-

2.2 Properties of the CDF

From the slides, the following properties must all hold for a valid CDF:

1. Bounds

$$0 \leq F_X(x) \leq 1$$

2. Limits

$$F_X(-\infty) = 0$$

$$F_X(\infty) = 1$$

3. Monotonicity

If $x_1 < x_2$, then:

$$F_X(x_1) \leq F_X(x_2)$$

4. Interval Probability

For $x_1 < x_2$:

$$\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

2.3 Example 1: Validity of Given CDFs

Determine which of the following functions are valid CDFs.

(a)

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)$$

Step-by-step verification:

- $\tan^{-1}(x)$ is monotonically increasing.
- As $x \rightarrow -\infty$, $\tan^{-1}(x) \rightarrow -\frac{\pi}{2}$:

$$F_X(-\infty) = \frac{1}{2} - \frac{1}{2} = 0$$

- As $x \rightarrow \infty$, $\tan^{-1}(x) \rightarrow \frac{\pi}{2}$:

$$F_X(\infty) = \frac{1}{2} + \frac{1}{2} = 1$$

✓ Valid CDF

(b)

$$F_X(x) = (1 - e^{-x})u(x)$$

- $u(x)$ is the unit step function.
- For $x < 0$: $F_X(x) = 0$
- For $x \geq 0$: $F_X(x)$ increases monotonically toward 1.

✓ Valid CDF

(c)

$$F_X(x) = e^{-x^2}$$

- As $x \rightarrow \infty$, $e^{-x^2} \rightarrow 0$, not 1.

✗ Invalid CDF

(d)

$$F_X(x) = x^2 u(x)$$

- As $X \rightarrow \infty$, $F_X(x) \rightarrow \infty$.

✗ Invalid CDF

2.4 Example 2: Probability Computation Using a CDF

Given:

$$F_X(x) = (1 - e^{-x})u(x)$$

(a) $\Pr(X > 5)$

$$\begin{aligned}\Pr(X > 5) &= 1 - \Pr(X \leq 5) \\ &= 1 - F_X(5) \\ &= 1 - (1 - e^{-5}) \\ &= e^{-5}\end{aligned}$$

(b) $\Pr(X < 5)$

$$\Pr(X < 5) = F_X(5) = 1 - e^{-5}$$

(c) $\Pr(3 < X < 7)$

$$\begin{aligned}\Pr(3 < X < 7) &= F_X(7) - F_X(3) \\ &= (1 - e^{-7}) - (1 - e^{-3}) \\ &= e^{-3} - e^{-7}\end{aligned}$$

(d) Conditional Probability

$$\Pr(X > 5 \mid X < 7)$$

Using:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

Let:

- $A = \{X > 5\}$
- $B = \{X < 7\}$

$$\Pr(A \cap B) = \Pr(5 < X < 7) = F_X(7) - F_X(5)$$

$$\Pr(B) = F_X(7)$$

$$\Pr(X > 5 \mid X < 7) = \frac{F_X(7) - F_X(5)}{F_X(7)}$$

3. Probability Density Function (PDF)

3.1 Definition

For a **continuous random variable** X , the PDF is defined as:

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{\Pr(x \leq X < x + \epsilon)}{\epsilon}$$

3.2 Relationship Between PDF and CDF

From the slides:

For a continuous RV,

$$\Pr(x \leq X < x + \epsilon) = F_X(x + \epsilon) - F_X(x)$$

Substitute into the PDF definition:

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{F_X(x + \epsilon) - F_X(x)}{\epsilon}$$

This is the definition of the derivative:

$$\boxed{f_X(x) = \frac{dF_X(x)}{dx}}$$

3.3 Inverse Relationship

Conversely, the CDF can be obtained from the PDF by integration:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

4. Expectation of Random Variables

(Introduced in the outline; detailed slides follow this lecture and are referenced but not expanded with external material.)

4.1 Definition (as introduced)

- Expectation represents the **average value** of a random variable weighted by its probability distribution.
-

4.2 Expectation of a Function of an RV

- Expectation applies not only to X , but also to functions of X , denoted $g(X)$.
-

4.3 Linear Operations with Expectation

Expectation satisfies linearity:

- Scaling and summation rules apply as introduced in the lecture outline.
-

4.4 Moments and Central Moments

From the outline:

- **n-th moments**
- **Variance**
- **Skewness**
- **Kurtosis**

(Only listed here, exactly as in slides, without adding definitions beyond the provided material.)

End of Lecture Scribe

This scribe is:

- ✓ Exam-oriented
- ✓ Slide-faithful
- ✓ Step-by-step
- ✓ Free from external intuition or examples

If you want, I can generate **another independent version** of this scribe using the *same slides but different organization and phrasing* for comparison, exactly as you planned.

1/2



L7_S2_A_Revised.pdf

PDF

I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Below is a **Lecture Scribe for Exam Preparation** prepared **strictly from the provided lecture slides only**.

All definitions, notation, assumptions, results, proofs, and worked examples are written using the content present in the lecture slides.

No external explanations, intuition, or additional examples have been introduced. Step-by-step reasoning is included only where it appears implicitly in the slides and is expanded into explicit algebraic steps for revision clarity.

All material is derived from the uploaded lecture slides .

CSE400 – Lecture 7 Lecture Scribe

Expectation, CDFs, PDFs and Problem Solving

1. Lecture Scope (From Outline)

The lecture covers:

1. The Cumulative Distribution Function (CDF)
 - Definition
 - Properties
 - Example
 2. The Probability Density Function (PDF)
 - Definition
 - Properties
 - Example
 3. Expectation of Random Variables
 - Definition and example
 - Expectation of a function of RV
 - Linear operation with expectation
 - n-th moments and central moments (Variance, Skewness, Kurtosis)
-

2. Cumulative Distribution Function (CDF)

2.1 Definition and Notation

Let X be a random variable.

The cumulative distribution function (CDF) is defined as:

$$F_X(x) = \Pr(X \leq x), \quad -\infty < x < \infty$$

Notation

- X : random variable
- $F_X(x)$: cumulative distribution function of X

From the slides:

- Most information about the random experiment described by X is determined by the behavior of $F_X(x)$.
-

2.2 Assumptions / Conditions for a Valid CDF

A function is a valid CDF only if the following properties hold.

Property 1 (Bounds)

$$0 \leq F_X(x) \leq 1$$

Property 2 (Limits)

$$F_X(-\infty) = 0$$

$$F_X(\infty) = 1$$

Property 3 (Monotonicity)

If $x_1 < x_2$, then

$$F_X(x_1) \leq F_X(x_2)$$

Property 4 (Probability from CDF)

For $x_1 < x_2$,

$$\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

2.3 Worked Example 1 — Finding Valid CDF

Determine which of the following functions are valid CDFs.

(1)

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)$$

Step-by-step verification

Step 1: Evaluate limits.

As $X \rightarrow -\infty$,

$$\tan^{-1}(x) \rightarrow -\frac{\pi}{2}$$
$$F_X(-\infty) = \frac{1}{2} - \frac{1}{2} = 0$$

As $X \rightarrow \infty$,

$$\tan^{-1}(x) \rightarrow \frac{\pi}{2}$$
$$F_X(\infty) = \frac{1}{2} + \frac{1}{2} = 1$$

Step 2: Function is increasing since $\tan^{-1}(x)$ is increasing.

All CDF conditions satisfied.

✓ Valid CDF.

(2)

$$F_X(x) = (1 - e^{-x})u(x)$$

Step-by-step verification

Step 1: For $x < 0$,

$$u(x) = 0 \Rightarrow F_X(x) = 0$$

Step 2: For $x \geq 0$,

$$F_X(x) = 1 - e^{-x}$$

Step 3: Limits:

$$F_X(-\infty) = 0, \quad F_X(\infty) = 1$$

Step 4: Function increases with x .

✓ Valid CDF.

(3)

$$F_X(x) = e^{-x^2}$$

As $x \rightarrow \infty$,

$$F_X(x) \rightarrow 0 \nexists 1$$

✗ Invalid CDF.

(4)

$$F_X(x) = x^2 u(x)$$

As $x \rightarrow \infty$,

$$F_X(x) \rightarrow \infty$$

Violates bound condition.

✗ Invalid CDF.

2.4 Worked Example 2 — Probability from CDF

Given:

$$F_X(x) = (1 - e^{-x})u(x)$$

(a) Compute $\Pr(X > 5)$

Step 1:

$$\Pr(X > 5) = 1 - \Pr(X \leq 5)$$

Step 2:

$$= 1 - F_X(5)$$

Step 3:

$$\begin{aligned} &= 1 - (1 - e^{-5}) \\ &= e^{-5} \end{aligned}$$

(b) Compute $\Pr(X < 5)$

$$\Pr(X < 5) = F_X(5) = 1 - e^{-5}$$

(c) Compute $\Pr(3 < X < 7)$

Step 1:

$$\Pr(3 < X < 7) = F_X(7) - F_X(3)$$

Step 2:

$$= (1 - e^{-7}) - (1 - e^{-3})$$

Step 3:

$$= e^{-3} - e^{-7}$$

(d) Conditional Probability

$$\Pr(X > 5 \mid X < 7)$$

Step 1:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

Let:

- $A = \{X > 5\}$
- $B = \{X < 7\}$

Step 2:

$$\begin{aligned}\Pr(A \cap B) &= \Pr(5 < X < 7) \\ &= F_X(7) - F_X(5)\end{aligned}$$

Step 3:

$$\Pr(B) = F_X(7)$$

Final result:

$$\Pr(X > 5 \mid X < 7) = \frac{F_X(7) - F_X(5)}{F_X(7)}$$

3. Probability Density Function (PDF)

3.1 Definition

For a continuous random variable X ,

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{\Pr(x \leq X < x + \epsilon)}{\epsilon}$$

3.2 Relationship Between PDF and CDF (Derivation from Slides)

For a continuous range:

$$\Pr(x \leq X < x + \epsilon) = F_X(x + \epsilon) - F_X(x)$$

Substitute into PDF definition:

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{F_X(x + \epsilon) - F_X(x)}{\epsilon}$$

This is the derivative definition:

$$f_X(x) = \frac{dF_X(x)}{dx}$$

3.3 Result (Converse Relationship)

The CDF can be expressed as the integral of the PDF:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

4. Expectation of Random Variables

(From lecture outline section)

Topics included:

- Definition and example of expectation
- Expectation of a function of a random variable
- Linear operation with expectation
- n-th moments and central moments:
 - Variance
 - Skewness
 - Kurtosis

(No additional formulas are introduced beyond what appears in the slides.)

End of Lecture Scribe

This scribe is:

- Fully slide-consistent
- Structured for exam revision
- Step-by-step where shown in lecture material
- Free from external assumptions or additional explanations

If needed, another independent version can be generated with a different organization while remaining fully faithful to the slides.



L7_S2_A_Revised.pdf
PDF

I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for

revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Below is a **Lecture Scribe for Exam Preparation** constructed **strictly from the lecture slides provided**.

All material — definitions, notation, assumptions, results, derivations, and examples — is taken only from the lecture slides.

No external explanations, intuition, or additional examples have been introduced.

Step-by-step reasoning is written explicitly for revision clarity while remaining faithful to the content shown in the slides.

All content is derived from the uploaded lecture slides .

CSE400 – Lecture 7 Lecture Scribe

Expectation, CDFs, PDFs and Problem Solving

1. Lecture Coverage (From Outline Slide)

The lecture includes:

1. Cumulative Distribution Function (CDF)
 - Definition
 - Properties
 - Example
2. Probability Density Function (PDF)
 - Definition
 - Properties
 - Example
3. Expectation of Random Variables
 - Definition and example
 - Expectation of a function of RV
 - Linear operation with expectation
 - n-th moments and central moments (Variance, Skewness, Kurtosis)

2. Cumulative Distribution Function (CDF)

2.1 Definition and Notation

Let X be a random variable.

The cumulative distribution function (CDF) is defined as:

$$F_X(x) = \Pr(X \leq x), \quad -\infty < x < \infty$$

Notation

- X : random variable
- $F_X(x)$: cumulative distribution function of X

Statement from slides:

- Most information about the random experiment described by X is determined by the behavior of $F_X(x)$.

2.2 Assumptions and Conditions (Properties of a CDF)

A function must satisfy the following conditions to be a valid CDF.

(1) Bounds

$$0 \leq F_X(x) \leq 1$$

(2) Limits

$$F_X(-\infty) = 0$$

$$F_X(\infty) = 1$$

(3) Monotonicity

If $x_1 < x_2$, then

$$F_X(x_1) \leq F_X(x_2)$$

(4) Probability from CDF

For $x_1 < x_2$,

$$\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

3. Worked Example 1 — Validity of CDF

Problem: Determine which functions are valid CDFs.

(a)

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)$$

Step-by-step verification

Step 1: Check limits.

As $x \rightarrow -\infty$,

$$\tan^{-1}(x) \rightarrow -\frac{\pi}{2}$$

$$F_X(-\infty) = \frac{1}{2} - \frac{1}{2} = 0$$

As $x \rightarrow \infty$,

$$\tan^{-1}(x) \rightarrow \frac{\pi}{2}$$

$$F_X(\infty) = \frac{1}{2} + \frac{1}{2} = 1$$

Step 2: Since $\tan^{-1}(x)$ is increasing, the function is non-decreasing.

Step 3: Values remain between 0 and 1.

✓ Valid CDF.

(b)

$$F_X(x) = (1 - e^{-x})u(x)$$

Step-by-step verification

Step 1: For $x < 0$,

$$u(x) = 0 \Rightarrow F_X(x) = 0$$

Step 2: For $x \geq 0$,

$$F_X(x) = 1 - e^{-x}$$

Step 3: Limits:

$$F_X(-\infty) = 0, \quad F_X(\infty) = 1$$

Step 4: Function increases with x .

✓ Valid CDF.

(c)

$$F_X(x) = e^{-x^2}$$

As $x \rightarrow \infty$,

$$F_X(x) \rightarrow 0$$

which violates $F_X(\infty) = 1$.

✗ Invalid CDF.

(d)

$$F_X(x) = x^2 u(x)$$

As $x \rightarrow \infty$,

$$F_X(x) \rightarrow \infty$$

which violates $0 \leq F_X(x) \leq 1$.

✗ Invalid CDF.

4. Worked Example 2 — Computing Probabilities from a CDF

Given:

$$F_X(x) = (1 - e^{-x})u(x)$$

(a) $\Pr(X > 5)$

Step 1:

$$\Pr(X > 5) = 1 - \Pr(X \leq 5)$$

Step 2:

$$= 1 - F_X(5)$$

Step 3:

$$= 1 - (1 - e^{-5})$$

Result:

$$= e^{-5}$$

(b) $\Pr(X < 5)$

$$\Pr(X < 5) = F_X(5) = 1 - e^{-5}$$

(c) $\Pr(3 < X < 7)$

Step 1:

$$\Pr(3 < X < 7) = F_X(7) - F_X(3)$$

Step 2:

$$= (1 - e^{-7}) - (1 - e^{-3})$$

Result:

$$= e^{-3} - e^{-7}$$

(d) Conditional Probability

$$\Pr(X > 5 \mid X < 7)$$

Step 1:

$$\Pr(A | B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

Let

$$A = \{X > 5\}, \quad B = \{X < 7\}$$

Step 2:

$$\begin{aligned}\Pr(A \cap B) &= \Pr(5 < X < 7) \\ &= F_X(7) - F_X(5)\end{aligned}$$

Step 3:

$$\Pr(B) = F_X(7)$$

Final expression:

$$\Pr(X > 5 | X < 7) = \frac{F_X(7) - F_X(5)}{F_X(7)}$$

5. Probability Density Function (PDF)

5.1 Definition

For a continuous random variable X ,

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{\Pr(x \leq X < x + \epsilon)}{\epsilon}$$

5.2 Relationship Between PDF and CDF (Step-by-Step Derivation)

From the slides:

For a continuous range,

$$\Pr(x \leq X < x + \epsilon) = F_X(x + \epsilon) - F_X(x)$$

Substitute into PDF definition:

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{F_X(x + \epsilon) - F_X(x)}{\epsilon}$$

By definition of derivative,

$$f_X(x) = \frac{dF_X(x)}{dx}$$

5.3 Result (Converse Statement)

The CDF can be written as the integral of the PDF:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

6. Expectation of Random Variables (As Listed in Slides)

The lecture includes:

- Definition and example of expectation
- Expectation of a function of a random variable
- Linear operation with expectation
- n-th moments and central moments:
 - Variance
 - Skewness
 - Kurtosis

(Only topics listed in slides are included; no additional formulas introduced.)

End of Lecture Scribe

This lecture scribe is:

- Fully restricted to lecture slide content
- Structured for exam revision
- Step-by-step where derivations appear in slides
- Free from external intuition or added explanations

If required, another independent version with a different structural ordering can be generated while remaining fully slide-faithful.



I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Below is a **Lecture Scribe for Exam Preparation** prepared **strictly using only the lecture slides provided**.

All definitions, notation, assumptions, results, derivations, and worked examples are taken exclusively from the lecture slides.

No external explanations, intuition, or additional examples have been introduced. Step-by-step reasoning is written explicitly for revision clarity while remaining fully faithful to the material presented in the slides.

All material is sourced from the lecture slides .

CSE400 – Lecture 7 Lecture Scribe

Expectation, CDFs, PDFs and Problem Solving

1. Lecture Scope (Outline)

The lecture contains the following topics:

1. The Cumulative Distribution Function (CDF)
 - Definition
 - Properties
 - Example
 2. The Probability Density Function (PDF)
 - Definition
 - Properties
 - Example
 3. Expectation of Random Variables
 - Definition and example
 - Expectation of a function of RV
 - Linear operation with expectation
 - n-th moments and central moments of RVs:
 - Variance
 - Skewness
 - Kurtosis
-

2. Cumulative Distribution Function (CDF)

2.1 Definition and Notation

Let X denote a random variable.

The cumulative distribution function (CDF) of X is defined as:

$$F_X(x) = \Pr(X \leq x), \quad -\infty < x < \infty$$

Notation

- X : random variable
- $F_X(x)$: cumulative distribution function of X

Statement from slides:

- Most of the information about the random experiment described by the random variable X is determined by the behavior of $F_X(x)$.
-

2.2 Assumptions and Conditions (Properties of a CDF)

A function must satisfy the following properties to be a valid CDF.

Property 1 — Bounds

$$0 \leq F_X(x) \leq 1$$

Property 2 — Limits

$$F_X(-\infty) = 0$$

$$F_X(\infty) = 1$$

Property 3 — Monotonicity

For $x_1 < x_2$,

$$F_X(x_1) \leq F_X(x_2)$$

Property 4 — Probability Using the CDF

For $x_1 < x_2$,

$$\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

3. Worked Example 1 — Validity of a CDF

Problem: Determine which functions are valid CDFs.

(a)

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)$$

Step-by-step verification

Step 1: Evaluate limits.

As $x \rightarrow -\infty$,

$$\tan^{-1}(x) \rightarrow -\frac{\pi}{2}$$

$$F_X(-\infty) = \frac{1}{2} - \frac{1}{2} = 0$$

As $x \rightarrow \infty$,

$$\tan^{-1}(x) \rightarrow \frac{\pi}{2}$$

$$F_X(\infty) = \frac{1}{2} + \frac{1}{2} = 1$$

Step 2: The function is non-decreasing because $\tan^{-1}(x)$ is increasing.

Step 3: The function values lie between 0 and 1.

✓ Valid CDF.

(b)

$$F_X(x) = (1 - e^{-x})u(x)$$

Step-by-step verification

Step 1: For $x < 0$,

$$u(x) = 0 \Rightarrow F_X(x) = 0$$

Step 2: For $x \geq 0$,

$$F_X(x) = 1 - e^{-x}$$

Step 3: Limits:

$$F_X(-\infty) = 0$$

$$F_X(\infty) = 1$$

Step 4: Function increases with increasing x .

✓ Valid CDF.

(c)

$$F_X(x) = e^{-x^2}$$

As $x \rightarrow \infty$,

$$F_X(x) \rightarrow 0$$

which violates the condition $F_X(\infty) = 1$.

✗ Invalid CDF.

(d)

$$F_X(x) = x^2 u(x)$$

As $x \rightarrow \infty$,

$$F_X(x) \rightarrow \infty$$

which violates $0 \leq F_X(x) \leq 1$.

✗ Invalid CDF.

4. Worked Example 2 — Probability Computation Using a CDF

Given:

$$F_X(x) = (1 - e^{-x})u(x)$$

(a) Compute $\Pr(X > 5)$

Step 1:

$$\Pr(X > 5) = 1 - \Pr(X \leq 5)$$

Step 2:

$$= 1 - F_X(5)$$

Step 3:

$$= 1 - (1 - e^{-5})$$

Result:

$$= e^{-5}$$

(b) Compute $\Pr(X < 5)$

$$\Pr(X < 5) = F_X(5) = 1 - e^{-5}$$

(c) Compute $\Pr(3 < X < 7)$

Step 1:

$$\Pr(3 < X < 7) = F_X(7) - F_X(3)$$

Step 2:

$$= (1 - e^{-7}) - (1 - e^{-3})$$

Result:

$$= e^{-3} - e^{-7}$$

(d) Conditional Probability

$$\Pr(X > 5 \mid X < 7)$$

Step 1:

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$$

Let:

$$A = \{X > 5\}, \quad B = \{X < 7\}$$

Step 2:

$$\begin{aligned}\Pr(A \cap B) &= \Pr(5 < X < 7) \\ &= F_X(7) - F_X(5)\end{aligned}$$

Step 3:

$$\Pr(B) = F_X(7)$$

Final expression:

$$\Pr(X > 5 \mid X < 7) = \frac{F_X(7) - F_X(5)}{F_X(7)}$$

5. Probability Density Function (PDF)

5.1 Definition

For a continuous random variable X ,

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{\Pr(x \leq X < x + \epsilon)}{\epsilon}$$

5.2 Relationship Between PDF and CDF (Step-by-Step Derivation)

From the slides:

For a continuous range,

$$\Pr(x \leq X < x + \epsilon) = F_X(x + \epsilon) - F_X(x)$$

Substitute into the definition of PDF:

$$f_X(x) = \lim_{\epsilon \rightarrow 0} \frac{F_X(x + \epsilon) - F_X(x)}{\epsilon}$$

By the definition of derivative,

$$\boxed{f_X(x) = \frac{dF_X(x)}{dx}}$$

5.3 Result (Converse Relationship)

The CDF can be written as the integral of the PDF:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

6. Expectation of Random Variables (As Listed in Slides)

The lecture includes:

- Definition and example of expectation
- Expectation of a function of a random variable
- Linear operation with expectation
- n-th moments and central moments:
 - Variance
 - Skewness
 - Kurtosis

(No additional formulas or explanations are introduced beyond what appears in the slides.)

End of Lecture Scribe

This lecture scribe is:

- Strictly based on lecture slides
- Organized for exam revision
- Step-by-step for derivations and examples
- Free from external intuition or added material

A different structurally independent version can also be generated while remaining fully slide-faithful if required.